# MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
*(A constituent unit of MAHE, Manipal)*

| DEPARTMENT OF Aeronautical and Automobile Engineering | | | |
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| **END SEMESTER EXAMINATION** | | **NOVEMBER,2024** | |
| **Level** | **B.Tech** | **Curriculum** | **2018** |
| **Program Name** | **Automobile Engineering** | **Course Type** | **PE** |
| **Course Name** | **Noise Vibration & Harshsness** | **Course Code** | **AAE 4037** |
| **Semester** | **VII** | **Max Marks** | **50** |
| **Date** | **29/11/2024** | **Duration** | **3 HOURS** |

**Instructions:**

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

| Q. No. | Question | Marks | NBA | | IET | | BL |
|---|---|---|---|---|---|---|---|
| | | | CO | PO | CLO | AHEP4 LO | |
| **1A** | Consider a narrow tube in which the source of sound is at one end and the longitudinal wave is moving towards the free end from the source as a 1-dimensional wave. Derive a relation for the acoustic natural frequencies of the narrow tube. The relationship between the particle velocity and the dynamic pressure is given by $\frac{\partial p}{\partial x} = -\rho_0 \frac{\partial u}{\partial x}$ Where $p$ is the dynamic pressure and $u$ is the particle velocity and $\rho_0$ is the density of the gas. Consider the harmonic wave given by phi= $a$ cos($wt$-$kx$) as the forward going wave | **04** | 1 | 1,2 | 1 | **1,2,7,18** | **03** |
| **1B** | A builder had acquired permission to construct an apartment in a residential area. There was a dilapidated building was there on the site which was quite huge. To start the construction, the old building had to be demolished and the demolition squad due to the deadline started using jack hammers and at once there were 10 labourers using it continuously. If each jack hammer produces approximately 100 dB, what is the total level produced by usage of 10 jack hammers at once? Is the builder breaking the law if in industrial areas, the permissible limit is 75 dB for daytime (07:00 to 22:00) and 70 dB at night (22:00 to 07:00). In commercial areas, it is 65 dB and 55 dB, while in residential areas it is 55 dB and 45 dB during daytime and night respectively. Consider incoherent acoustic source. | **04** | 1 | 1,2 | 1 | **1,2,7,18** | **04** |

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| **1C** | Assess the reasons for choosing Sound power level to rate the noise levels of different machineries rather than sound pressure or intensity. | **02** | 2 | 1,2 | 2 | **1,2,7,18** | **03** |
| **2A** | Discuss the working principle of a moving coil microphone with relevant sketch. If given a choice between choosing condenser microphone and moving coil microphone, which one would you choose if you had to conduct acoustic characterization of a newly synthesized material to be used a sound insulating barrier? | **04** | 2 | 1,2 | 2 | **1,2,7,18** | **02** |
| **2B** | Discuss about time averaging in a sound level meter and what is the difference between fast, slow impulse and peak measurements? | **03** | 2 | 1,2 | 2 | **1,2,7,18** | **02** |
| **2C** | What is the difference between pressure field and diffuse filed with respect to an omnidirectional spherical acoustic source? If given a choice of calibrating a microphone which type of filed is preferable and what is the necessity of calibration? | **03** | 1 | 1,2 | 1 | **1,2,7,18** | **02** |
| **3A** | Consider that you have been given the task of designing the task of designing the intake system of a passenger car. Consider a snorkel position at the top of the front grille, above the radiator. The filter box is to be mounted on the inner wheel arch. This effectively fixes the snorkel length to, for example, 375 mm. What are the factors deciding the snorkel cross section are and the dimensions of the filter box? | **03** | 3 | 1,2,3,7 | 3 | **1,2,7,13,18** | **04** |
| **3B** | In a typical automobile, the intake system is the dominating source of noise during a pass-by noise test when the vehicle is approaching the test area. Discuss the sources of intake noise and differentiate between primary, secondary noise sources and the shell noise. Analyse the method to differentiate the primary and secondary noise generation mechanism | **04** | 3 | 1,2,3,7 | 3 | **1,2,7,13,18** | **03** |
| **3C** | Basic intake and exhaust systems are made up of expansions, contractions and pipe protrusions. Analyse the influence of these three types of designs on the flow duct acoustics and how do these influence the noise created by the vehicle? | **03** | 3 | 1,2,3,7 | 3 | **1,2,7,13,18** | **04** |
| **4A** | Adopting a suitable acoustic prediction method during the development programme can help reduce programme time and cost by minimising the number of prototypes. Analyse the differences between frequency domain and time domain prediction methods. | **03** | 3 | 1,2,3,7 | 3 | **1,2,7,13,18** | **03** |
| **4B** | In case of exhaust noise control, most of the techniques involved in the case of intake system also are applicable to exhaust system design. But there many notable differences between the two. Analyse how the exhaust system differs from the intake. | **04** | 4 | 1,2,3,7 | 4 | **1,2,13,18** | **03** |
| **4C** | Discuss how the use of porous asphalt can influence the tyre noise generation and what are the limitations of that? | **03** | 4 | 1,2,3,7 | 4 | **1,2,13,18** | **03** |

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| **5A** | Consider a typical passenger car with dimensions of 3mx1.6m x 1.3m (*lbh*). Compute the (1, 0, 0), (0, 1, 0), (2, 0, 0), (0, 0, 1), (1, 1, 0), (2, 1, 0) acoustic modes of the car. Assume the speed of sound as 350 m/sec. | **05** | 5 | 1,2,3,7 | 5 | **1,2,13,18** | **04** |
| **5B** | Differntiate between subjective and objective assessment of interior noise of a typical passenger car. | **03** | 5 | 1,2,3,7 | 5 | **1,2,13,18** | **02** |
| **5C** | Analyse the engine noise mechanism with respect to a passenger car and discuss the difference between combustion noise and mechanical noise. | **02** | 5 | 1,2,3,7 | 5 | **1,2,13,18** | **03** |

## MARK DISTRIBUTION (CO-wise & BTL-wise)

| CO / CLO | Total Marks |
|---|---|
| **1** | ***11*** |
| **2** | ***9*** |
| **3** | ***13*** |
| **4** | ***7*** |
| **5** | ***10*** |

| BL | Total Marks |
|---|---|
| **2** | **13** |
| **3** | **22** |
| **4** | **15** |
| **5** | |